Roll No. .....

BBA–303

B. B. A. (THIRD SEMESTER)

MID SEMESTER EXAMINATION, Oct., 2022

STATISTICS FOR MANAGEMENT

Time : 1½ Hours

Maximum Marks : 50

Note : (i) Answer all the questions by choosing any ***one*** of the sub-questions.

(ii) Each sub-question carries 10 marks.

1. (a) "Statistics are numerical statements of facts, but all facts numerically stated are not statistics." Comment upon the statement. (CO1/CO2)

OR

(b) Find Mean, Median and Mode from the following distribution : (CO1/CO2)

Class	Frequency
30—40	6
40—50	10
50—60	16
60—70	14
70—80	10
80—90	5
90—100	2

P. T. O.

2. (a) Discuss briefly the applications of business statistics in business decision-making. Support your answer with examples. (CO2/CO3)

OR

(b) Compute coefficient of variation for factory A and B from the following : (CO2/CO3)

Daily Wages	Workers (Factory A)	Workers (Factory B)
12	5	15
15	10	25
17	20	30
22	35	35
25	12	21
30	8	19

Find :

(i) Which of the following factory pays higher average daily wages ?

(ii) In which factory are wages more variable ?

3. (a) Discuss briefly the meaning of correlation and its types with examples. (CO2/CO3)

OR

(b) Calculate the coefficient of correlation by Karl Pearson's method from the following data relating to overhead expenses and the cost of production : (CO2/CO3)

Overhead Expenses	Cost
80	15
90	15
100	16
110	19
120	17
130	18
140	16
150	18
160	19

4. (a) "Skewness can be defined as lack of symmetry." Illustrate. (CO3/CO4)

OR

(b) From the following data, calculate Karl Pearson's coefficient of skewness : (CO3/CO4)

Marks (above)	No. of Students
0	150
10	140
20	100
30	80
40	80
50	70
60	30
70	14
80	0

5. (a) Differentiate between correlation and regression analysis. Write the mathematical properties of coefficient of correlation. (CO1/CO4)

OR

(b) From the data given below, find : (CO1/CO4)

Marks of Economics	Marks of Statistics
25	43
28	46
35	49
32	41
31	36
36	32
29	31
38	30
34	33
32	39

(i) The two regression equations (Y on X and X on Y)

(ii) The most likely marks in Statistics when marks in Economics are 35.